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BIOLOGY

0610/63

Paper 6 Alternative to Practical

October/November 2025

1 hour

You must answer on the question paper.

No additional materials are needed.

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- You should show all your working and use appropriate units.

INFORMATION

- The total mark for this paper is 40.
- The number of marks for each question or part question is shown in brackets [].

This document has **12** pages. Any blank pages are indicated.

- 1 Yeast cells produce an enzyme called invertase. Invertase catalyses the breakdown of sucrose into reducing sugars. Sucrose is **not** a reducing sugar.

A student investigated the effect of two different concentrations of invertase extract on a sucrose solution.

The student used this method:

Step 1 Label three test-tubes **A**, **B** and **W**.

Step 2 Use a syringe to put 2 cm^3 of sucrose solution into all three test-tubes.

Step 3 Use a clean syringe to put 2 cm^3 of 100% invertase extract into test-tube **A**.

Step 4 Use a clean syringe to put 2 cm^3 of 10% invertase extract into test-tube **B**.

Step 5 Use a clean syringe to put 2 cm^3 of distilled water into test-tube **W**.

Step 6 Put all three test-tubes in a warm water-bath at 40°C for 15 minutes.

Step 7 Measure the temperature of the liquid in the test-tubes.

(a) State the name of the piece of equipment that is used to measure temperature.

..... [1]

Step 8 Use a clean syringe to add 4 cm^3 of Benedict's solution to all three test-tubes.

Step 9 Place all three test-tubes into a hot water-bath for five minutes.

Step 10 After five minutes, record the colour of the liquid in each test-tube.

Fig. 1.1 shows the notes the student made about their observations in step 10.

	the liquid in test-tube A is brick-red
	the liquid in test-tube B is green
	W remains blue

Fig. 1.1



(b) Prepare a table and record the results of this investigation using the information in Fig. 1.1.

[3]

(c) State a conclusion for the results shown in test-tube **A** and test-tube **B**.

.....
.....
..... [1]

(d) Explain the purpose of test-tube **W**.

.....
.....
.....
..... [2]

(e) State the independent variable in this investigation.

..... [1]

(f) Explain why the test-tubes were placed in the warm water-bath in step 6.

.....
.....
..... [1]





(g) Explain why a clean syringe was used in steps 3, 4 and 5.

.....

.....

..... [1]

(h) Invertase is a protein.

State the reagent used to test for the presence of protein.

..... [1]

[Total: 11]



- 2 A student investigated the effect of temperature on the rate of respiration in yeast cells.

They used the apparatus shown in Fig. 2.1.

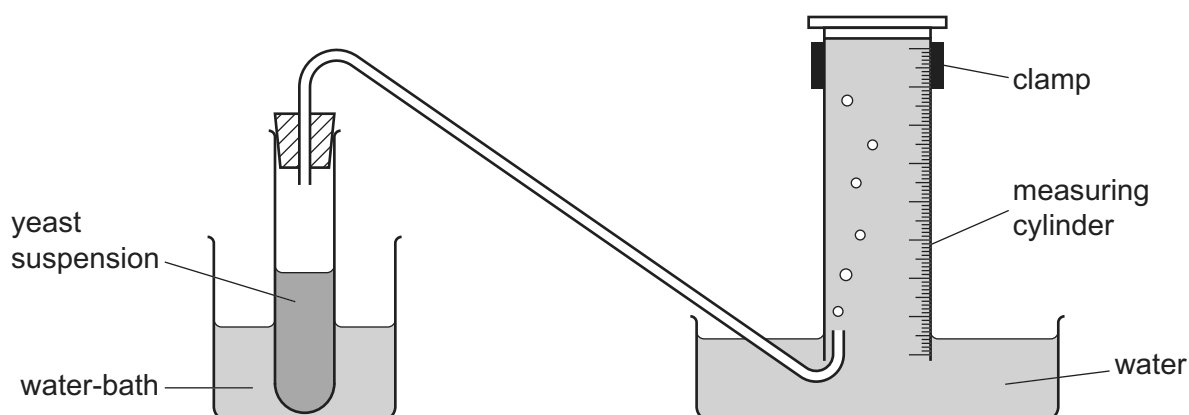


Fig. 2.1

The student used this method:

- Put 25 cm³ of yeast suspension into a test-tube.
- Put the test-tube into a water-bath and add water at 10 °C to the water-bath.
- Leave the apparatus for five minutes before moving on to the next step.
- Start a stop-clock and count the number of bubbles produced by the respiring yeast cells in three minutes.
- Repeat the procedure using a water-bath maintained at different temperatures.

- (a) (i) State **two** variables that the student kept constant in this investigation.

1

.....

2

.....

[2]

- (ii) State **one** possible source of error when measuring the dependent variable **and** suggest an improvement to this method of measuring.

error

.....

improvement

.....

.....

[2]



(b) Table 2.1 shows the results of the investigation.

Table 2.1

water-bath temperature /°C	number of bubbles produced in 3 minutes	rate of bubble production /bubbles per minute
10	33	11
20	45	15
30	79	
40	112	37
50	124	41
60	37	12

(i) Calculate the rate of bubble production at 30 °C.

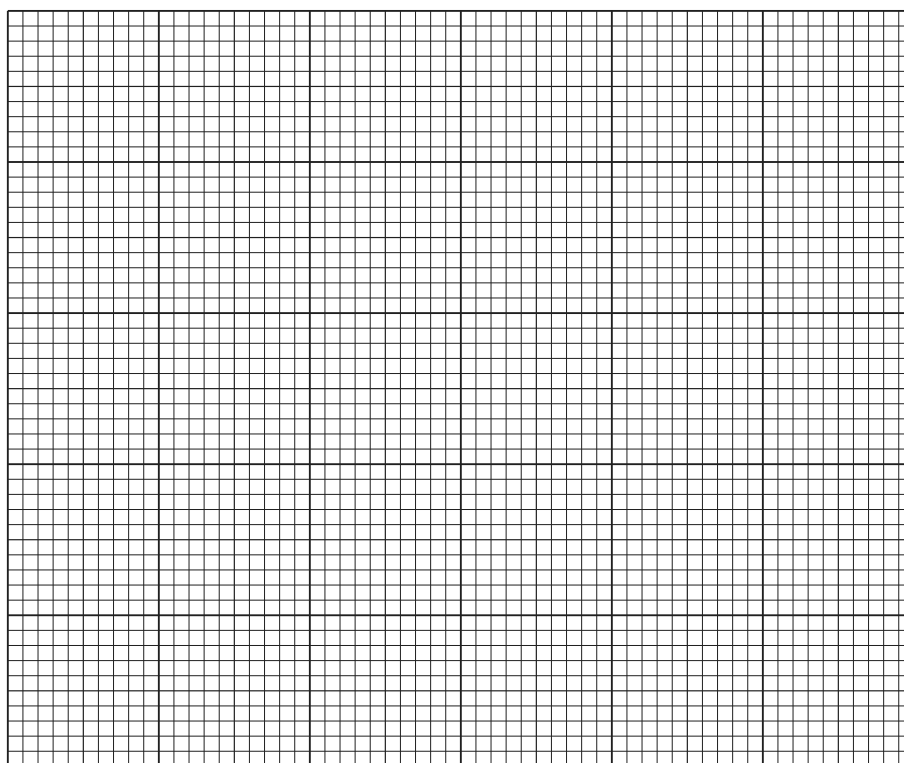
Give your answer to an appropriate number of significant figures.

Space for working.

..... bubbles per minute [2]



- (ii) Plot the data in Table 2.1 on the grid to show the effect of temperature on the **rate** of bubble production.



[4]

- (iii) Describe the results of this investigation.

.....

.....

.....

.....

..... [2]

- (c) Describe how the student could confirm that the gas produced is carbon dioxide.

.....

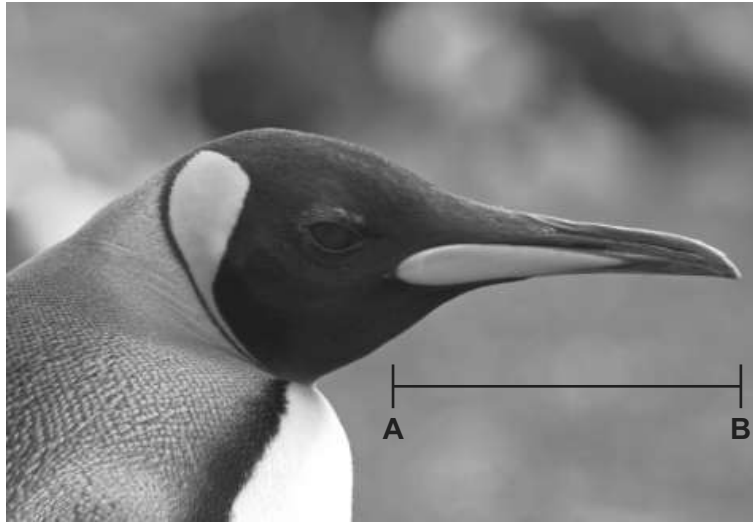
.....

..... [2]

[Total: 14]



- 3 (a) Fig. 3.1 is a photograph of the head of a king penguin.



magnification $\times 0.42$

Fig. 3.1

- (i) Make a large drawing of the head of the king penguin shown in Fig. 3.1.



(ii) The length of line **AB** represents the length of the penguin's beak.

Measure the length of line **AB** in Fig. 3.1.

length of line **AB** mm

Use your measurement and the formula to calculate the actual length of the penguin's beak.

$$\text{magnification} = \frac{\text{length of line } \mathbf{AB} \text{ in Fig. 3.1}}{\text{actual length of the penguin's beak}}$$

Give your answer to **one** decimal place.

Space for working.

..... mm
[3]



(b) Fig. 3.2 shows the heads of a gentoo penguin and a black-footed penguin.



gentoo



black-footed

not to scale

Fig. 3.2

Other than size, state **two** ways the gentoo penguin differs from the black-footed penguin in Fig. 3.2.

- 1
- 2

[2]





- (c) Penguins are covered in feathers. The feathers reduce heat loss by insulating the birds from the cold.

A beaker of hot water can be used to represent a penguin.

Plan an investigation to determine the effect of the thickness of an insulating material on heat loss in a beaker of hot water.

[6]

[Total: 15]





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